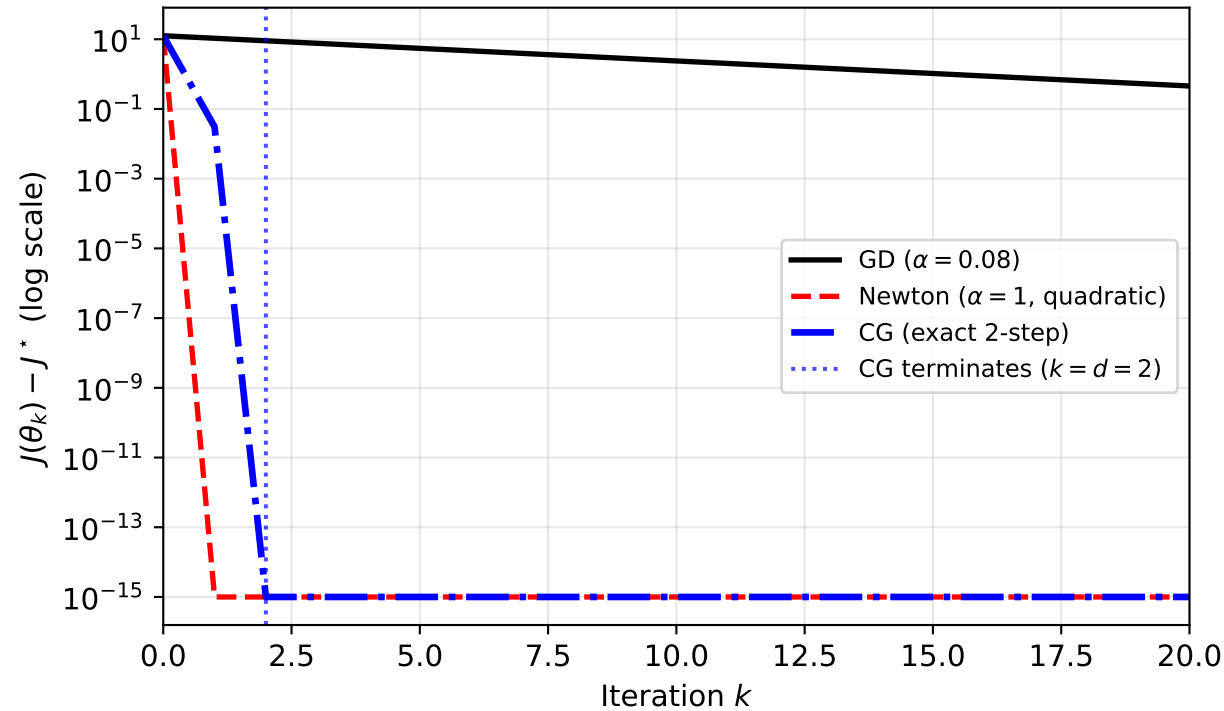


Conjugate Gradient: Exact 2-Step Termination on 2D Quadratic Benchmark A
(Newton = 1 step; GD = 80+ steps to match CG accuracy)

GD vs Newton vs CG on Benchmark A
(Quadratic Loss, $d = 2$, $\kappa \approx 10$)



With Theoretical Convergence Bounds

